Sumit Yadav

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SKILLS

Languages:	Python, Java, C/C++, JavaScript, SQL
AI/ML:	Machine Learning, NLP, RAG, Computer Vision, Semantic Search, Embeddings, Similarity Search, XGBoost, Deep Learning
Frameworks & Libraries:	FastAPI, PyTorch, Scikit-learn, Transformers, Streamlit, NumPy, Pandas
Databases & Tools:	MySQL, PostgreSQL, FAISS, Git, GitHub, Linux
Core CS Fundamentals:	Data Structures & Algorithms (DSA), Database Management System (DBMS), Operating Systems (OS), Computer Networks (CN), Object-Oriented Programming (OOPs)
Soft Skills:	Problem Solving, Critical Thinking, Adaptability, Leadership

PROJECTS

Landcover-segmentation | *Python, PyTorch, U-Net, DeepLabV3+, ResNet-34, Computer Vision* *Nov' 26*

- Developed a semantic segmentation pipeline using **PyTorch**, implementing **U-Net** and **DeepLabV3+** with pretrained **ResNet-34** to classify satellite imagery into five land-cover classes.
- Implemented **data augmentation**, **class-weighted Cross-Entropy** and **Dice Loss** to address class imbalance and evaluated models using **mIoU**, **F1-score** and **Pixel Accuracy**.
- Built a **Streamlit** inference application featuring model selection, color-coded segmentation overlays, land-cover analytics and downloadable prediction masks.

ClaimShield — Insurance Claim Analysis | *Python, PyTorch, YOLOv8, Transformers, RAG, FastAPI* *Aug' 26*

- Built a multimodal AI system combining **NLP**, **Computer Vision**, and **document retrieval** for automated insurance claim analysis.
- Developed pipelines to analyze claim descriptions and damage images, identifying inconsistencies between reported information and visual evidence.
- Integrated **RAG-based policy retrieval** and **FastAPI APIs** to support automated claim coverage analysis and application integration.

AI Digital Asset Manager | *Python, CLIP, Sentence Transformers, FAISS, NLP, Streamlit, CrossEncoder* *Apr' 26*

- Built an AI-powered document management system implementing **RAG with FAISS**, vector embeddings and **Groq LLMs** for context-aware document question answering.
- Developed semantic and hybrid search with intelligent **chunking**, **similarity retrieval**, conversational memory and source citations for document-specific responses.
- Integrated **multimodal AI** using **OpenAI CLIP** for zero-shot image classification and **Tesseract OCR** for text extraction, with an interactive Streamlit interface.

TRAINING

Summer PEP Training 2026 – CP 2028 | Programming Pathshala *Jul' 26*

- **Competitive Programming:** Strengthened problem-solving and algorithmic thinking through structured CP training.
- **Data Structures & Algorithms:** Practiced core DSA concepts and efficient algorithmic approaches for programming problems.
- **Problem Solving:** Developed skills in analyzing problems, optimizing solutions, and applying programming fundamentals.

CERTIFICATES

Oracle Certified Foundations Associate Agentic AI | ***Oracle*** *Sep' 26*
MOOC on REACT.js | ***TechVeda*** *Mar' 25*
Basic to Beyond Python | ***CsePathshala*** *Jan' 25*

ACHIEVEMENTS

- **Finalist, Dev Storm Hackathon** *Aug' 26*
  Advanced to the final round by developing and presenting a competitive technical solution.
- **150+ LeetCode Problems Solved** *Aug' 25*
  Solved across core DSA topics, including arrays, strings, trees, graphs, and DP.
- **Coordinator, Student Academic Advisory Committee (CSE Department)** *Jan' 25*
  Helped restructure and modernise the Computer Science curriculum by providing recommendations on relevant technologies, courses, and industry-oriented skills.

EDUCATION

Lovely Professional University *Phagwara, Punjab*
Bachelor of Technology in Computer Science and Engineering; CGPA: **8.74** *Jul' 24 – Present*
Army Public School *Prayagraj, Uttar Pradesh*
Intermediate: **87.8%** (2023) | Matriculation: **92.4%** (2021) *Mar' 20 – May' 23*